Name: A. Tabasiya

Batch: 2 to 4

EC: 6815

TEST - 2

(1.) Write the characteristics of list with necessary code.

Characteristics of list:-

1. Different datatypes - yes
2. Dynamic Memory Allocation - yes
3. Duplicate element - yes
4. Mutable or immutable - Mutable
5. Ordered or unordered - Ordered
6. Slicing supported - yes
7. Index supported - yes
8. Sorting - NO

Example program.

```
Elements: [10,20,30,40,50,60,70,80]
print ("listElements:" elements)
data = [10,20,30.,"python"]
print ("Duplicate elements:", elements)
print ("Ordered :" elements)
print ("Index :" elements[0])
print ("slicing :", element[1:4])
element [0] = 100
print ("After modification:", elements)
```

(2) Chasactesistics of Tuple with neccessay code.

(1) Dymanic memosy allocation - NO
(2) Diffesant datatype - yes
(3) Duplicates - yes
(4) mutable os Immutable - Immutable
(5) osdesed os unosdesed - osdesed
(6) Index supposted - yes Supposted
(7) Sliing is supposted - yes
(8) Sosting - NO

Example psogsam:-

```
elements = (90, 80, 70, 60, 50, 40, 30, 20, 10)
psint (" Tuple element :", elements)
data = (10, 20, 30, "python")
psint (" Diffesat elements :", elements)
psint (" osdesed element :", elements)
psint ("Index5 :", element [5])
psint (" sliing :", [2:6])
psint (" Sosted elemat :" Sosted (elements))
```

(3) Chasactesistic of Set:-

(1) Dyramic memosy allocatio - yes
(2) Diffesent datatype - yes
(3) Duplicates - NO
(4) Mutable os Immutable - Mutable
(5) osdesed os unosdesed - mutable unosded
(6) Index supposted - NO

⑦ Slicing supported - NO.

⑧ Sorting - NO

(4) Example program:

```
Student = {"name": "Tabas", "age": 21,
           "city": "Chennai", "roll number": 10}
print("Student detail:", Student)
print("Different Data Types:" Student)
Student["age"] = 21
print("Student details:", Student)
Student["city"] = "Madurai"
print("After changing city:", Student)
```

(4.) Looping statement:-

Looping statements are used to execute a block of code repeatedly, Two looping statement are these ; i.) for loop & ii.) While loop.

Sample program.

→ (print 1 to 5 using a for loop.)

```
for i in range(1, 6):
    print(i)
```

→ (Star pattern using for loop

```
for i in range(1, 6):
    print("*" * i)
```

While loop:-

```
count = 1
while count <= 8:
    print(count)
    count += 1
```

O/p:- 1, 2, ---- 8

⑤ Chasactesistics of Dict:-

① Dynamic memosy allocatio - yes
② Diffesent datatypes - yes
③ Duplicates - key - no values - yes
④ Mutable or Immutable - mutable
⑤ Osdesed or unOsdesed - osdesed
⑥ Index suppested or - NO
⑦ Slicing suppested - NO

Example psogsam:-

```
products = {"Rice": 60, "Sugas": 40,
            "oil": 120, "milk": 20}

print(products)
print("Rice price:", products["Rice"])
products["Sugas"] = 50
print("Updated products:", products).

products = {"Rice": 60, "Sugas": 45,
            "Rice": 75}
Print("Duplicate key":(products))
```